

Project: Escape Room Puzzle Game

Phase: Week 7 (Riddles, Scoring & Mini-Map)

1. Objective

This phase extends the escape room game with a guided riddle system, a scoring mechanism based on player performance, and quality-of-life features (mini-map and pause) to improve navigation and playability.

2. Detailed Task Breakdown

2.1 Clue Scroll Riddles (5 total)

- A new ClueScroll entity type is added alongside regular items. Each scroll is placed at a fixed location in the room.
- Collecting a scroll does not fill inventory space; instead it immediately reveals a short riddle text in a dialog/toast on screen.
- Each of the 5 riddles gives a textual hint pointing toward the location or identity of the treasure/next required item (e.g. directional clues, object descriptions).
- A RiddleManager class stores all 5 riddles in order and tracks which ones have been revealed (revealedCount), so the game knows how many clues the player has already read.
- Riddles are revealed strictly in sequence the player cannot skip ahead to riddle 3 before reading riddles 1 and 2.

2.2 Scoring System

- A ScoreManager class tracks two performance metrics:
 - coinsCollected — incremented each time the player picks up a coin item.
 - elapsedTimeMillis — measured from game start to the moment the player wins.
- Final score formula:

$$\text{score} = (\text{coinsCollected} \times \text{POINTS_PER_COIN}) + \text{TIME_BONUS}$$

$$\text{TIME_BONUS} = \max(0, (\text{TIME_LIMIT} - \text{elapsedTime}) / 1000) \times \text{TIME_BONUS_MULTIPLIER}$$

- POINTS_PER_COIN (e.g. 50 points per coin) rewards exploration.
- TIME_BONUS rewards finishing quickly the faster the player escapes, the higher the bonus.
- The score is calculated once, at the moment the Victory condition is triggered, and displayed on the end-game overlay.

2.3 Mini-Map Toggle (M key)

- Pressing **M** toggles a boolean flag `showMiniMap`.
- When active, a small scaled-down overlay is drawn in a corner of the screen showing:
 - The outline of the current room.
 - The player's current position as a dot.
 - Positions of uncollected items, scrolls, and the door/treasure.
- The mini-map is purely visual and does not pause gameplay — movement and timers continue normally while it is shown.

2.4 Pause Functionality (ESC key)

- Pressing **ESC** toggles the game state to PAUSED (extending the existing GameState enum: PLAYING, PAUSED, VICTORY, GAME_OVER).
- While paused:
 - The game loop's tick() method returns immediately (no movement, no timer countdown, no item collection).
 - A semi-transparent "Paused" overlay is drawn over the frozen frame.
 - Pressing ESC again resumes play from exactly where it left off.

4. Deliverables

- ☐ Coin-and-time-based scoring system implemented
- ☐ Mini-map toggle (M) implemented
- ☐ Pause functionality (ESC) implemented
- ☐ Final score displayed correctly on victory
- ☐ Tested full playthrough with riddles, scoring, mini-map, and pause together

5. Tools & Resources

- Java JDK 11+
- IntelliJ IDEA / VS Code

- Java Swing & AWT Documentation
- OOP Concepts (state management, encapsulation)
- Existing Escape Room Game codebase